RESEARCH PROPOSAL PRESENTATION

Feature Drift Detection Techniques via Adversarial Validation

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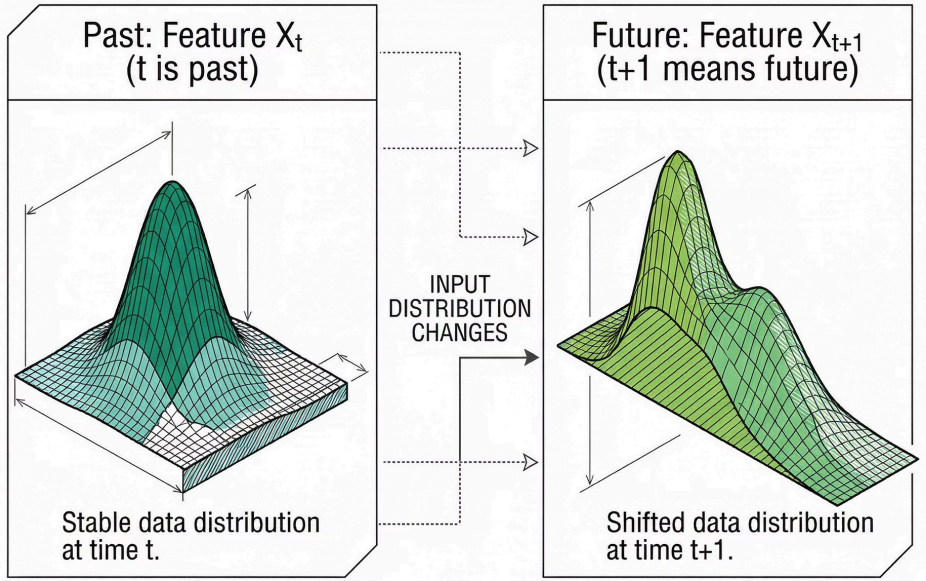


Table of Contents

1. **Introduction**
2. **Literature Review**
3. **Limitations & Related Work**
4. **Methodology**
5. **Preliminary Results**
6. **Research Focus**
7. **Conclusion**
8. **References**

Introduction

Why Drift Matters in Production ML

Common Sources of Drift

- **User behavior changes:** Shifting preferences, seasonal patterns, emerging trends
- **Environmental factors:** Economic conditions, regulatory changes, competitive landscape
- **Policy updates:** Business rule modifications, pricing adjustments, product launches
- **Data pipeline changes:** Sensor recalibration, logging modifications, upstream system updates

Key Insight: Drift signals often emerge *before* accuracy metrics decline, providing a critical window for intervention.

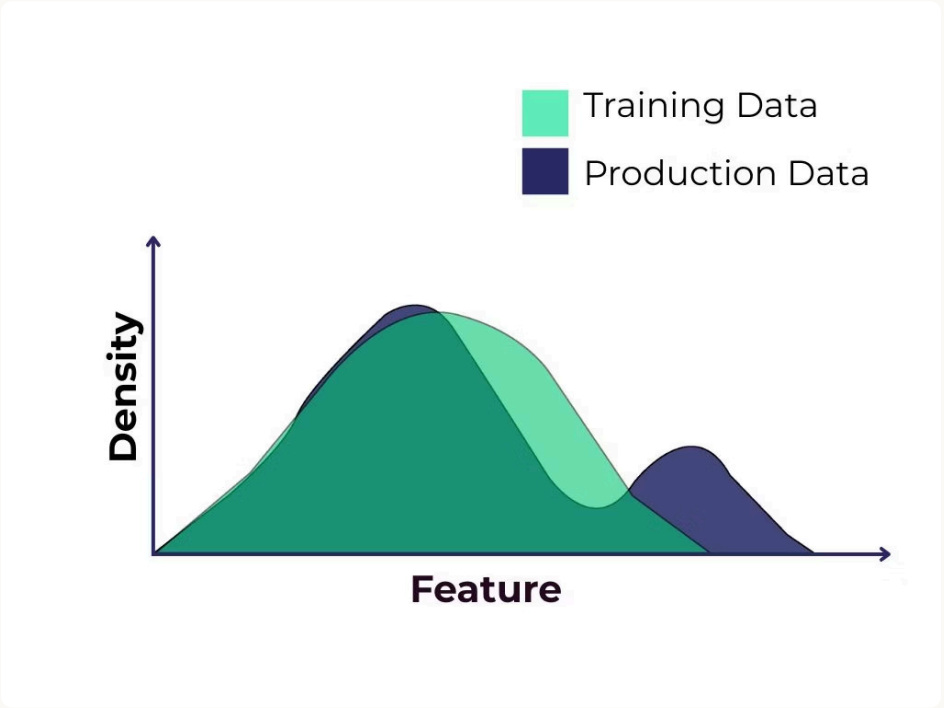


Figure: Drift in production

References: Sculley et al. (2015) Hidden Technical Debt in Machine Learning Systems; Breck et al. (2017) ML Test Score for Production Readiness

Literature Review

Core definitions and mathematical framework

Mathematical Framework

In a data stream at time *t*, we observe joint distribution $P_t(X, Y)$. Drift occurs when:

$$P_t(X, Y) \neq P_{t\pm1}(X, Y)$$

Given that, P_t is probability of feature(X) at a given time t, and Pt±1 is the changed probability over time.

X represents Features
Y represents Label

−1 (Past Perspective)
+1 (Future/Predictive Perspective)

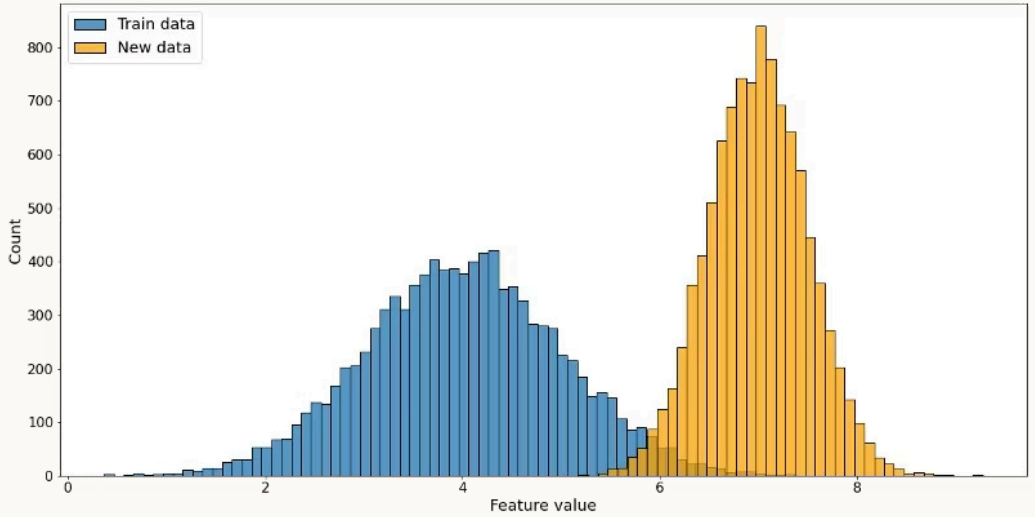
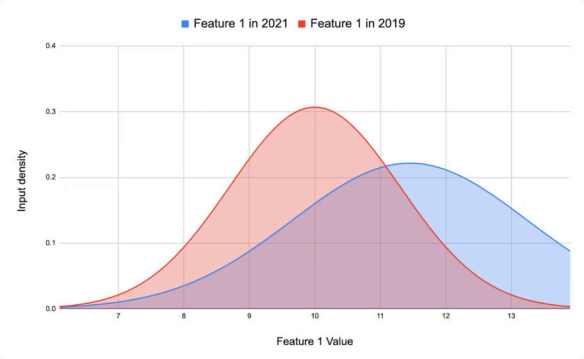
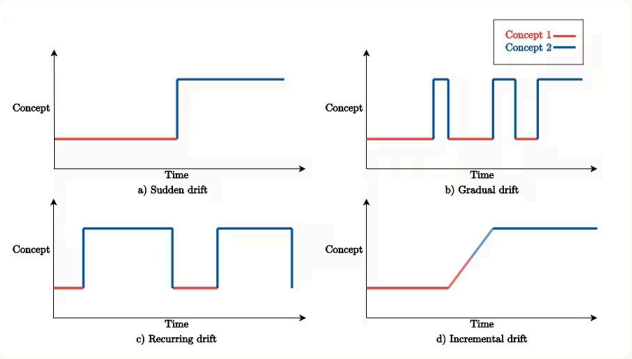


Figure: Graphs showing Data Drift from Geeksforgeeks

References: Lu et al. (2019), Gama et al. (2014), Tsymbal (2004), Shimodaira (2000)

Two Concepts of Drift





1. Drift by Nature of Change

Nature of Change (The "What" and "Where")

Nature of Change = **Component-based** (The *source* of the change).

- *Question:* "What part of the data actually changed?"

2. Drift by Temporal Pattern

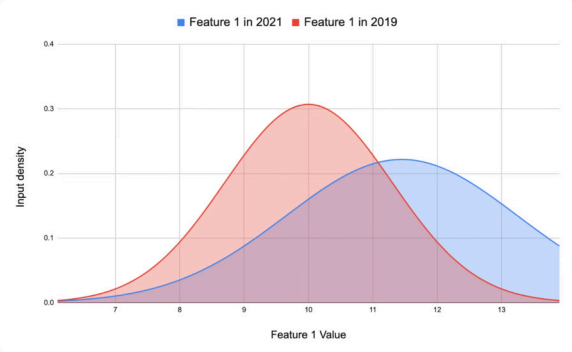
Temporal (The "When" and "How Fast")

Temporal = **Time-based** (The *timeline* of the change).

- *Question:* "How fast did it happen?"

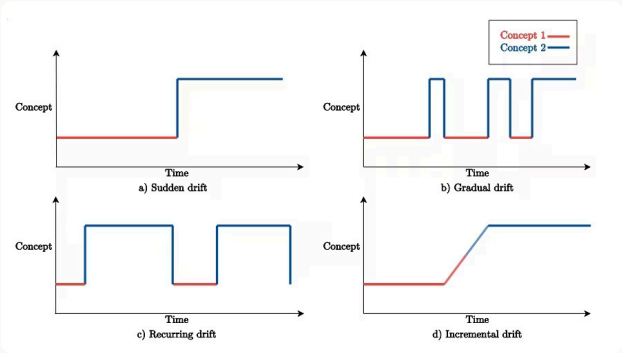
> **Terminology Ambiguity:** Be aware that "Concept Drift" is sometimes used loosely. Gama et al. (2014) use it as an **umbrella term** for *any* distribution change, whereas Tsymbal (2004) restricts it strictly to changes in P(y|X).

Two Concepts of Drift – Continued



A) Drift by Nature of Change

1. **Concept Drift (Real Drift)**
2. **Covariate Drift (Feature Drift / Virtual Drift)**
3. **Prior Probability Drift (Label Shift)**



B) Drift by Temporal Pattern

1. **Abrupt (Sudden) Drift**
2. **Gradual Drift**
3. **Incremental Drift**
4. **Recurring (Seasonal) Drift**

> **Terminology Ambiguity:** Be aware that "Concept Drift" is sometimes used loosely. Gama et al. (2014) use it as an **umbrella term** for *any* distribution change, whereas Tsymbal (2004) restricts it strictly to changes in P(y|X).

02. LITERATURE REVIEW

(A) Drift by Nature of Change

Formal Definition of Drift Types (Lu et al., 2019; Gama et al., 2014), Given, *joint distribution P(X, y) where X = features, y = labels:*

Drift Types	Mathematical Definition	Detection Approach
1.Concept Drift (Real Drift)	$P_t(y\|X) \neq P_{t+1}(y\|X)$ Input-output relationship changes (X and Y Changes)	Requires labels
2.Covariate Drift (Feature Drift)	$P_t(X) \neq P_{t+1}(X)$ Input distribution changes (X changes)	Label-free
3.Prior Probability Drift (Label Shift)	$P_t(y) \neq P_{t+1}(y)$ Output/Class distribution changes (Y changes)	Requires labels

References: Lu et al. (2019), Gama et al. (2014), Tsymbal (2004), Shimodaira (2000)

02. LITERATURE REVIEW

Concept Drift (Real Drift)

Concept Drift occurs when the underlying relationship between input features and target labels changes over time, requiring models to adapt to new patterns.

Definition

The relationship between inputs X and outputs y changes over time.

Mathematical Formula

$$P_t(y|X) \neq P_{t\pm1}(y|X)$$

Example

The criteria for what constitutes a "fraudulent transaction" or "spam" email *evolve* as malicious actors adapt their tactics, making previous rules obsolete.

Detection

Requires **actual labels** to observe the changed input-output relationship, often after a model's performance has already degraded.

Covariate Drift (Feature Drift / Virtual Drift)

Covariate drift, also known as feature drift or virtual drift, occurs when the statistical properties of the input features change over time, even if the underlying relationship between inputs and outputs remains the same.

Definition

The distribution of input features (X) changes, but the relationship between inputs and outputs ($y|X$) remains constant.

Mathematical Formula

$$P_t(X) \neq P_{t\pm1}(X)$$

Example

A fraud detection model trained on transaction data primarily from **urban users** is later applied to data from **rural users**. The types of transactions (features) might differ, but the definition of what constitutes fraud remains consistent.

Detection

Can be detected **without requiring labels** for the new data, making it suitable for early warning systems. This aspect is a key focus of this thesis.

02. LITERATURE REVIEW

Prior Probability Drift (Label Shift)

Prior Probability Drift, also known as Label Shift, occurs when the overall prevalence of classes changes over time, even if the features within each class maintain their statistical distribution.

Definition

The class distribution $P(y)$ changes over time, while the conditional distribution of features given the class $P(X|y)$ remains stable.

Mathematical Formula

$$P_t(y) \neq P_{t\pm1}(y), \text{ but } P_t(X|y) = P_{t\pm1}(X|y)$$

Example

Consider a predictive model for flu outbreaks. During winter months, the proportion of positive flu cases (label 'flu') naturally increases, even if the symptoms (features X) for a flu patient remain the same.

Detection

Requires access to **ground truth labels** for the new data to accurately track the shifting proportions of each class.

(B) Temporal Pattern

Drift Tyoes	Mathematical Representation	Detection Approach
1.Abrupt (Sudden) Drift	$$\mathcal{D}_t = \begin{cases} \mathcal{D}_{source} & \text{if } t < T \\ \mathcal{D}_{target} & \text{if } t \geq T \end{cases}$$	Window-Based Statistical Tests
2.Gradual Drift	$$\mathcal{D}_t = (1 - v(t)) \cdot \mathcal{D}_{source} + v(t) \cdot \mathcal{D}_{target}$$ where $v(t)$ increases from 0 to 1	Distance-Based & Moving Averages
3.Incremental Drift	$$\mathcal{D}_t = P(X, y\|\theta_t) \quad \text{where} \quad \theta_{t+1} = \theta_t + \delta$$	Cumulative Error Monitoring
4.Recurring (Seasonal) Drift	$$\mathcal{D}_t \approx \mathcal{D}_{t \bmod k}$$ or $$\mathcal{D}_t = \mathcal{D}_{t+k}$$	Context-Aware / Hybrid Methods

References: Lu et al. (2019), Gama et al. (2014), Žliobaitė (2010)

Abrupt (Sudden) Drift

1 Definition

A rapid, discrete and immediate shift from one data distribution to a completely different one.

2 Mathematical Representation

$$P_t(X) \rightarrow P_{t+1}(X) \text{ instantaneously at time } t^*$$

3 Visualization

Often depicted as a **step function**, where the distribution sharply changes at a specific point in time.

4 Examples

Triggered by distinct events such as a **system migration**, a major **policy change**, or a sudden **market crash**.

5 Detection Implications

Requires a **fast response** from detection systems; generally **easier to detect** due to its clear and strong signal compared to other drift types.

Gradual Drift

1 Definition

The new distribution progressively replaces the old one over time, rather than a sudden shift.

2 Mathematical Representation

$$P_t(X) = (1 - \alpha_t) \cdot P_{old}(X) + \alpha_t \cdot P_{new}(X), \text{ where } \alpha_t \text{ increases from } 0 \to 1$$

3 Visualization

Depicted as a **smooth transition** between distributions, often resembling an S-curve or ramp over time.

4 Example

A **gradual shift in customer demographics** (e.g., increasing age or income) over several months due to evolving market trends or new product adoption.

5 Detection Challenge

The signal emerges **slowly and subtly**, making it challenging to distinguish from normal data noise or minor fluctuations without careful monitoring.

Incremental Drift

1 Definition

Small, continuous changes accumulate over time, leading to a significant shift in data distribution rather than a sudden event.

2 Mathematical Representation

$$P_{t+1}(X) = P_t(X) + \epsilon_t, \text{ where } \epsilon_t \text{ are small increments}$$

3 Visualization

Often depicted as a **creeping or fluctuating line**, showing a continuous trend rather than a sharp break or smooth transition.

4 Examples

Gradual sensor degradation in industrial equipment, **evolving user preferences** on a streaming service, or **slow shifts in socio-economic factors** affecting market models.

5 Detection Challenge

Each individual change is minute and subtle, making it hard to distinguish from normal data noise. The **cumulative effect** only becomes apparent over longer observation periods.

Recurring (Seasonal) Drift

1 Definition

Previously observed distributions reappear cyclically, often linked to predictable temporal patterns.

2 Mathematical Representation

$$P_t(X) = P_{t-T}(X), \text{ where } T \text{ is the period}$$

3 Visualization

Often depicted as an **oscillating pattern**, where data distributions transition between known states cyclically.

4 Examples

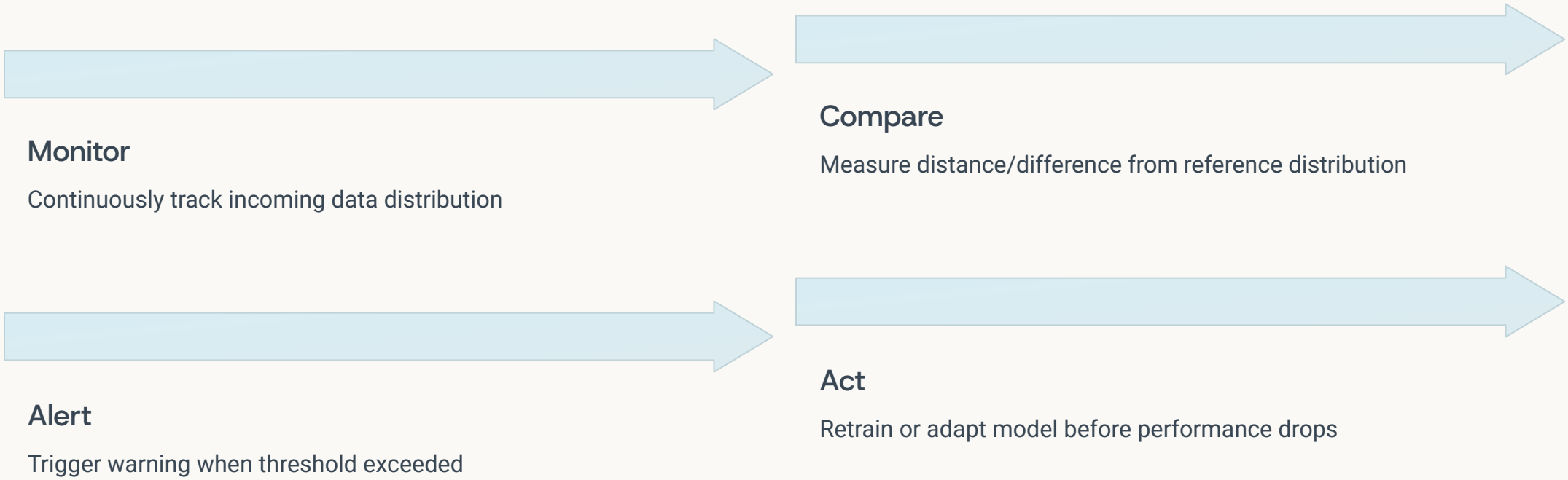
Typical instances include **holiday shopping spikes**, **seasonal demand fluctuations** (e.g., ice cream sales), or **annual disease outbreaks**.

5 Detection Challenge

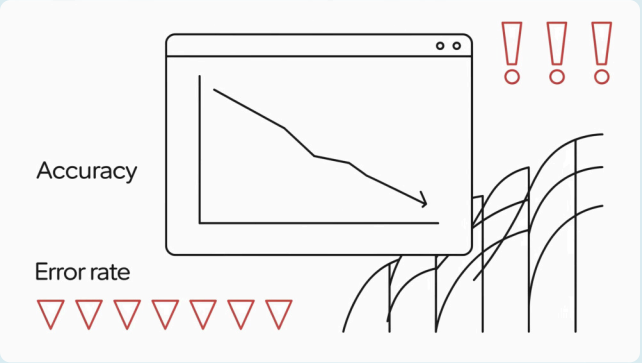
Requires mechanisms to **recognize and adapt** to known recurring patterns, distinguishing them from novel drift, and **avoiding false alarms**.

Drift detection is part of machine learning operations.

Drift Detection: Identifying when data distribution changes significantly
Goal: Signal drift BEFORE performance degrades



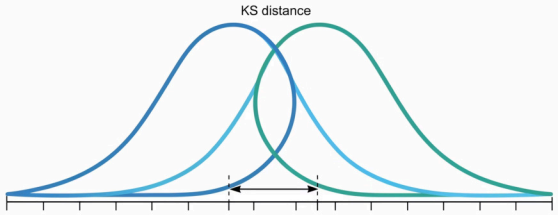
Drift Detection Methods



Supervised Drift Detection

Monitors model performance metrics (accuracy, error rate) over time. Requires ground-truth labels. Detects drift after performance degradation occurs.

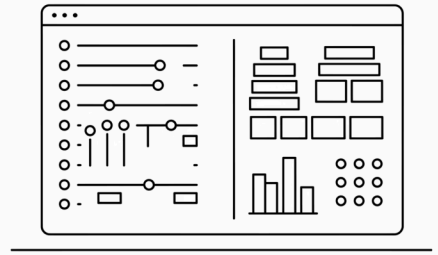
Examples: DDM, EDDM, ADWIN



Unsupervised Drift Detection

Compares feature distributions using statistical tests. No labels required. Enables early detection before performance drops.

Examples: KS Test, PSI, MMD, Chi-Square



Hybrid Drift Detection

Combines supervised and unsupervised approaches. Leverages strengths of both methods for robust detection.

Examples: Ensemble Methods, **Contexr Aware Drift (Cobb et al., 2022)**

Each approach has trade-offs: supervised methods are accurate but reactive, unsupervised methods are proactive but may have false alarms, and hybrid methods aim to balance both strengths.

Supervised Drift Detection Methods

Supervised drift detection leverages ground-truth labels to identify performance degradation over time, indicating a shift in the underlying data distribution.

Method Name	Description	Pros	Cons	Reference
DDM	Monitors the model's error rate increase.	Simple to implement, well-established method, provides statistical guarantees	Requires true labels, reactive (detects after performance drop).	Gama et al. (2004)
EDDM	Tracks the distance between consecutive errors.	Earlier detection than DDM, better for gradual drift	Label-dependent, sensitive to noise, aims for earlier detection.	Baena-García et al. (2006)
ADWIN	Adaptive sliding window that tracks changes in the average value of a random variable.	Automatically adapts window size, effective for concept drift and concept evolution, no need to set window size manually	Requires labels, reactive detection (after performance drop)	Bifet & Gavalda (2007)
HDDM	Utilizes Hoeffding's inequality to bound the error rate.	Robust against certain noise, provides theoretical guarantees via Hoeffding bounds	Needs labels, delayed detection, robust against certain noise.	Frías-Blanco et al. (2015)

02. LITERATURE REVIEW

Unsupervised Drift Detection Methods

These methods do not require ground-truth labels, enabling earlier detection of distribution shifts before model performance degrades.

Method	Description	Pros	Cons	Reference
KS Test	Compares Cumulative Distribution Functions (CDFs) of two samples.	Non-parametric, statistically sound, simple to implement.	Univariate only, misses multivariate shifts, sensitive to noise.	Massey (1951)
PSI	Population Stability Index, measures changes in feature distributions.	Widely used, simple to understand for univariate changes.	Univariate only, misses multivariate shifts, requires binning, threshold-dependent.	Siddiqi (2017), Yurdakul (2018), Lin (2017)
KSWIN	Kolmogorov-Smirnov Test with Sliding Window for streaming data.	Adapts KS test for streaming data, can detect changes in real-time.	Univariate, noise-sensitive, can have false positives, computationally intensive.	Raab et al. (2020)

02. LITERATURE REVIEW

Hybrid Drift Detection Methods

Hybrid methods combine the strengths of different detection techniques to achieve more robust and adaptive drift monitoring, aiming to overcome the limitations of individual approaches.

Method Name	Description	Pros	Cons	Key References
Ensemble Detection	Integrates multiple base detectors, often from different categories (e.g., statistical and supervised).	More resilient system.	Significantly increases complexity and the challenge of tuning numerous parameters effectively.	Krawczyk et al. (2017)
Adaptive Ensemble	Dynamically switches between different drift detection methods based on observed data characteristics or performance.	Dynamically adaptive and promising.	Introduces computational overhead and often lacks clear, robust criteria for method selection.	Gomes et al. (2017)

The Missing Piece: Limitations & Related Work

03. LIMITATIONS & RELATED WORK

Gap in Existing Methods:

Supervised Methods (DDM, EDDM)

Require ground-truth labels, leading to reactive rather than proactive drift detection.

Unsupervised Statistical Tests (KS, PSI)

Primarily univariate, failing to capture complex changes in joint distributions.

Hybrid Uncertainty-Based Methods

Heavily rely on model calibration, which can often be unreliable and difficult to maintain.

Our Contribution:



Adversarial Validation

A novel approach for **multivariate, label-free,** and **model-agnostic** drift detection.



Systematic Benchmark

Comprehensive evaluation framework for drift detection, addressing a significant gap in prior research.



Practical MLOps Guidance

Actionable recommendations for integrating robust drift detection into real-world ML production systems.

Adversarial Validation (AV): A Bridging Solution

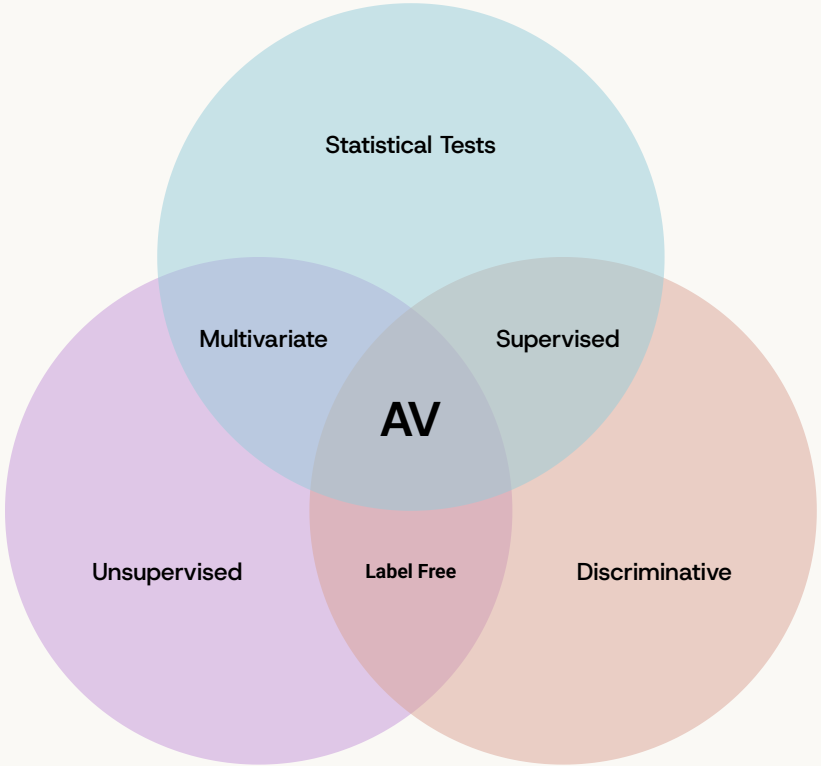


Figure: A Venn Diagram showing all features of Adversarial Validation (AV)

03. LIMITATIONS & RELATED WORK

Adversarial Validation: Origins & Applications

Adversarial Validation, though powerful, has primarily been applied in batch settings, revealing a significant gap in its use for continuous streaming data.

Uber's MaLTA System (Pan et al., 2020)

Utilized for intelligent batch sample selection before model retraining. Effectively detected train-deployment data distribution mismatches

Credit Scoring (Qian et al., 2021)

Applied adversarial classifiers to filter training samples, enhancing model robustness against distribution shifts in financial data.

Context-Aware Drift (Cobb & Van Looveren, 2022)

The ContextMMDDrift detector in Alibi Detect (SeldonIO) explicitly implements context-aware drift detection, allowing for conditional drift analysis (Cobb & Van Looveren, 2022).

Uber's MaLTA System (Pan et al., 2020)

Context

A production user-targeting system at Uber, monitoring dataset shift specifically before model retraining events.

Mechanism

Trains an adversarial classifier to differentiate between historical and current deployment data, using its performance as a drift signal.

Operational Use

Flags data mismatches to trigger targeted retraining or curated dataset selection, optimizing model update cycles.

Limitations

- Batch/periodic checks, not designed for real-time streaming data.
- High computational cost for repeated classifier training and permutation tests.
- Sensitivity to window sizing, class imbalance, and classifier selection, potentially causing false alarms.

03. LIMITATIONS & RELATED WORK

Credit Scoring (Qian et al., 2021)

Context

Addresses nonstationary applicant populations and regulatory risks prevalent in credit-scoring pipelines, where data distributions can change rapidly.

Mechanism

Utilizes adversarial validation to detect data shifts. It intelligently selects training subsamples that closely match the current deployment distribution for model retraining.

Impact

Significantly improves model fairness and robustness by ensuring retraining occurs on representative data. Also aids in identifying features that are leaked or time-dependent.

Limitations

- Primarily applied offline or periodically, not suited for low-latency, real-time streaming data environments.
- Subsampling introduces potential selection bias, impacting generalization.
- Privacy (PII) and regulatory constraints complicate automated and frequent retraining decisions.

Context-Aware Drift (Cobb & Van Looveren, 2022)

Context

Addresses situations where traditional drift detection often flags changes in P(x) as drift, even when these changes are explainable by shifts in context P(c). It aims to differentiate between true performance-impacting drift and acceptable, context-driven variations.

Mechanism

The ContextMMDDrift detector in Alibi Detect implements the Conditional Maximum Mean Discrepancy (CMMD) test. This statistical test compares how data x relates to context c in a reference set versus a test set, evaluating if $P(x \mid c)$ has truly shifted.

Operational Use

To use ContextMMDDrift, you must explicitly separate input data into features (x) – the data to monitor for anomalies, and context (c) – the data allowed to change. This separation is fundamental to its conditional analysis.

Limitations

- **Curse of Dimensionality (Context):** Performance degrades as the dimensionality of the context c increases.
- **Computational Cost:** CMMD tests can be computationally expensive, especially with high-dimensional x and c.
- **Data Requirements:** Requires sufficient data to accurately estimate conditional distributions, which can be challenging for rare contexts.

Methodology

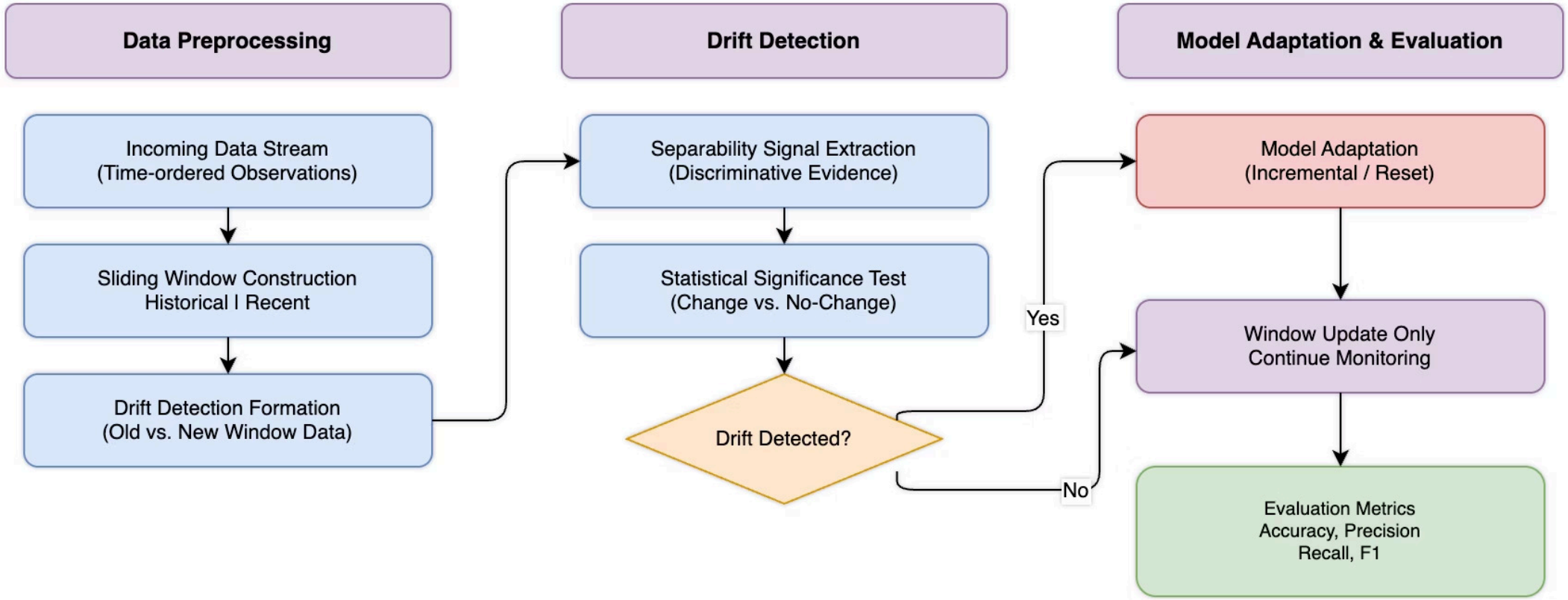


Figure: Overview of Proposed Methodology

Step 1: Collect Two Windows of Data

The initial step in our streaming drift detection methodology involves segmenting the continuous data stream into two distinct, non-overlapping windows: a historical **Reference Window (W_ref)** and a recent **Current Window (W_cur)**. This fundamental separation allows for a direct comparison between established data patterns and newly observed samples.

In Code:

```
X_ref = X[ref_start:ref_end]  # Historical samples
X_cur = X[cur_start:cur_end]  # Recent samples
```

Step 2: Create a Binary Classification Problem

The core of Adversarial Validation lies in reframing drift detection as a binary classification task. We transform the problem into: *"Can a classifier distinguish between samples from the historical reference data and the current incoming data?"*

In Code:

```
# Stack features from both windows
X = np.vstack([X_ref, X_cur])
y = np.concatenate([
    np.zeros(len(X_ref)),
    np.ones(len(X_cur))
])

#  # Assign label 0 to reference samples and Assign label 1 to current samples
```

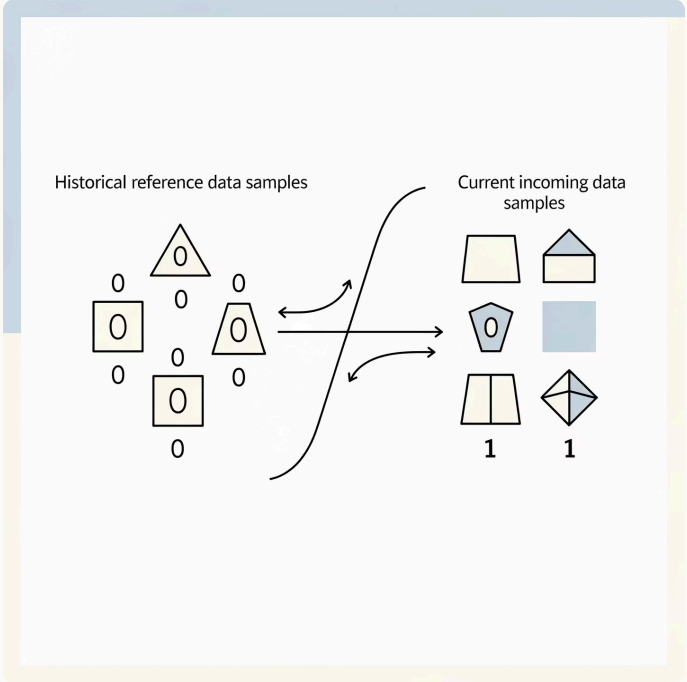


Figure: Example of label assign in samples

Step 3: Train a Classifier

The core of Adversarial Validation is to train a binary classifier ((Random Forest, Logistic Regression) to distinguish between samples from the historical `W_ref` and the current `W_cur` data windows. This model effectively learns the differences in distributions between the two datasets.

The classifier's task is simple: given a data sample, predict if it belongs to the reference (label 0) or current (label 1) window. This transforms the complex problem of drift detection into a straightforward classification problem.

In Code:

```
clf = RandomForestClassifier(n_estimators=100,
max_depth=5)
clf.fit(X_scaled, y)
```

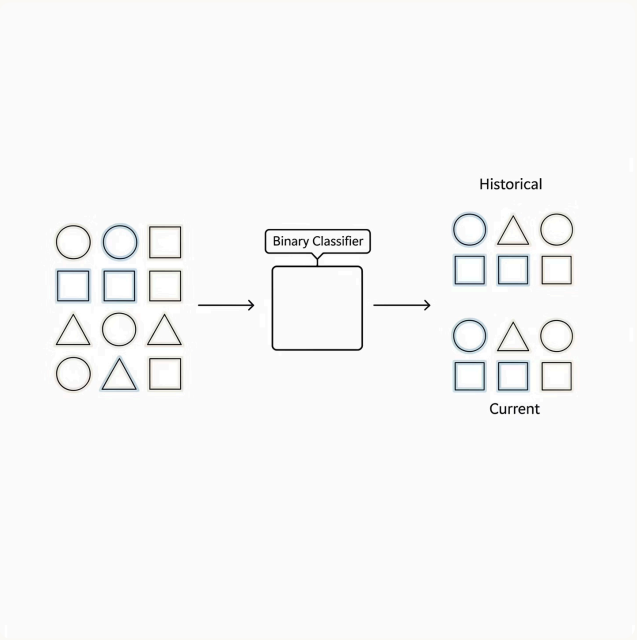


Figure: Example of ML model flow

Step 4: Measure Separability with AUC

AUC Value	Interpretation
0.50	Random guessing → Distributions are identical (no drift)
0.60-0.70	Slight separation → Possible drift
0.80+	Strong separation → Clear drift detected

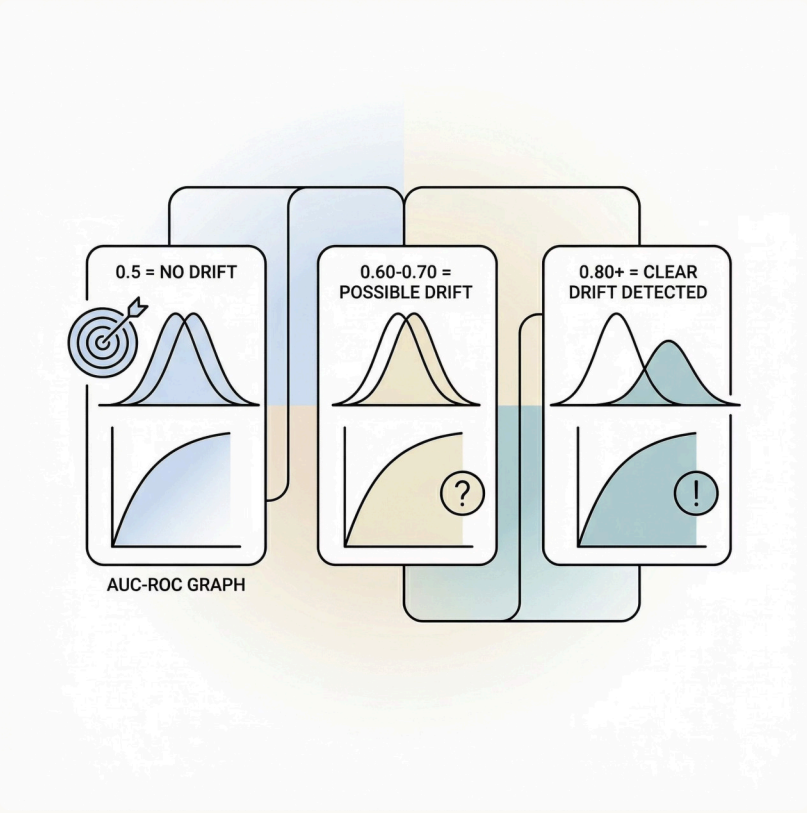


Figure: Threshold of traditional adversarial validation

Step 5: Statistical Significance via Permutation Test

1. Observe Actual AUC

Calculate the AUC using the trained classifier on the correctly labeled reference and current data.

2. Shuffle Labels

Randomly scramble the labels (0s for reference, 1s for current) multiple times, breaking any true association with the data windows.

3. Recompute "Null" AUC

For each shuffled set of labels, retrain the classifier and compute a "null" AUC.

4. Repeat & Compare

Repeat shuffling and recomputing many times to build a distribution of null AUCs, then compare the observed AUC against this distribution.

Drift is indicated when the **p-value < 0.05** and the **AUC ≥ 0.70**.

If the p-value ≥ 0.05 or AUC < 0.70, no significant drift is detected.

Formula:

The **p-value** indicates the probability of observing an AUC as extreme as, or more extreme than, the actual observed AUC by random chance:

$$p - value = \frac{\text{Number of null AUCs} \geq \text{observed AUC}}{B}$$

Preliminary Results

Experiment 1: Synthetic / Generated Dataset

Experiment 1A: Abrupt Drift (Synthetic Dataset)

This experiment validates the adversarial validation methodology using a synthetic dataset designed with an abrupt drift. We simulate a clear shift in data distribution for specific features and observe if our method can successfully detect it.

Experiment Configuration:

```
drift_point     5000
n_samples       10000
n_features      5
AV classifier   LogisticRegression + StandardScaler
class_weight    balanced
CV folds        5
Permutation B       200
alpha       0.05

Streaming window rule:
W_ref = X[t - 500 : t], W_cur = X[t : t + 500]
start t = 500, step = 200
```

Experiment 1A: Abrupt Drift (Synthetic Dataset)

Analysis of feature statistics confirms the expected abrupt drift in specific features, while others remain stable.

Feature	Before Drift (Mean +/- Std)	After Drift (Mean +/- Std)	Drift?
Feature_0	0.0098 +/- 1.0179	1.9902 +/- 1.0116	YES
Feature_1	0.0194 +/- 0.9936	2.0094 +/- 1.0005	YES
Feature_2	-0.0162 +/- 1.0020	-0.0049 +/- 1.0083	NO
Feature_3	-0.0166 +/- 0.9918	0.0066 +/- 0.9965	NO
Feature_4	0.0080 +/- 0.9864	-0.0100 +/- 0.9919	NO

The results demonstrate that Feature_0 and Feature_1 exhibit a clear increase in their mean values by approximately 2.0, precisely matching the engineered drift. This confirms the effectiveness of the synthetic dataset in simulating abrupt data distribution shifts.

Results

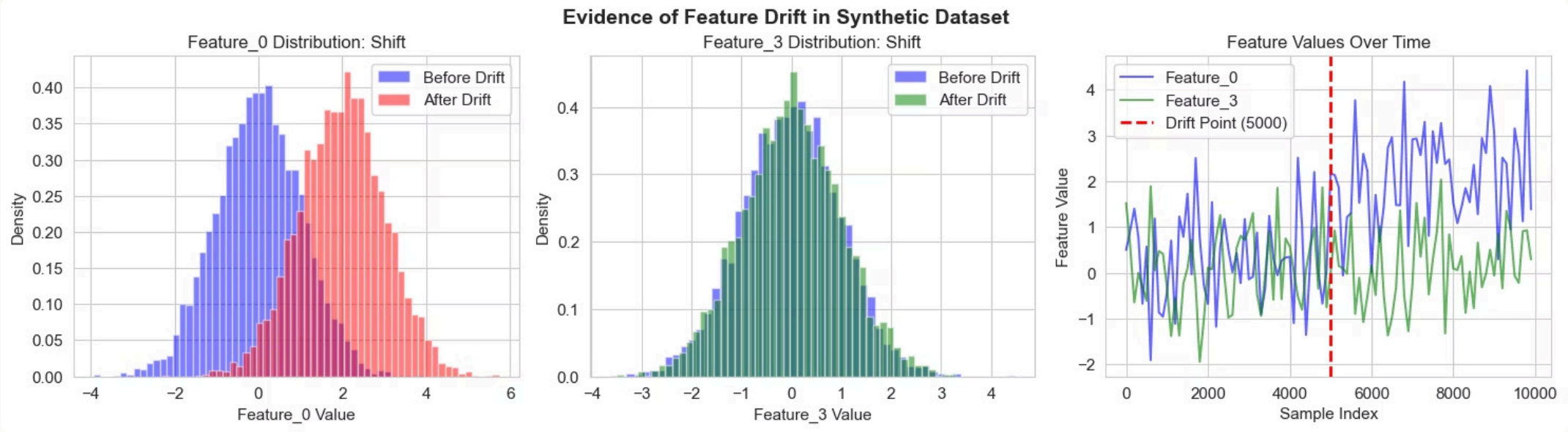


Figure: Histogram and Time-Series visualization of Abrupt Drift

Results

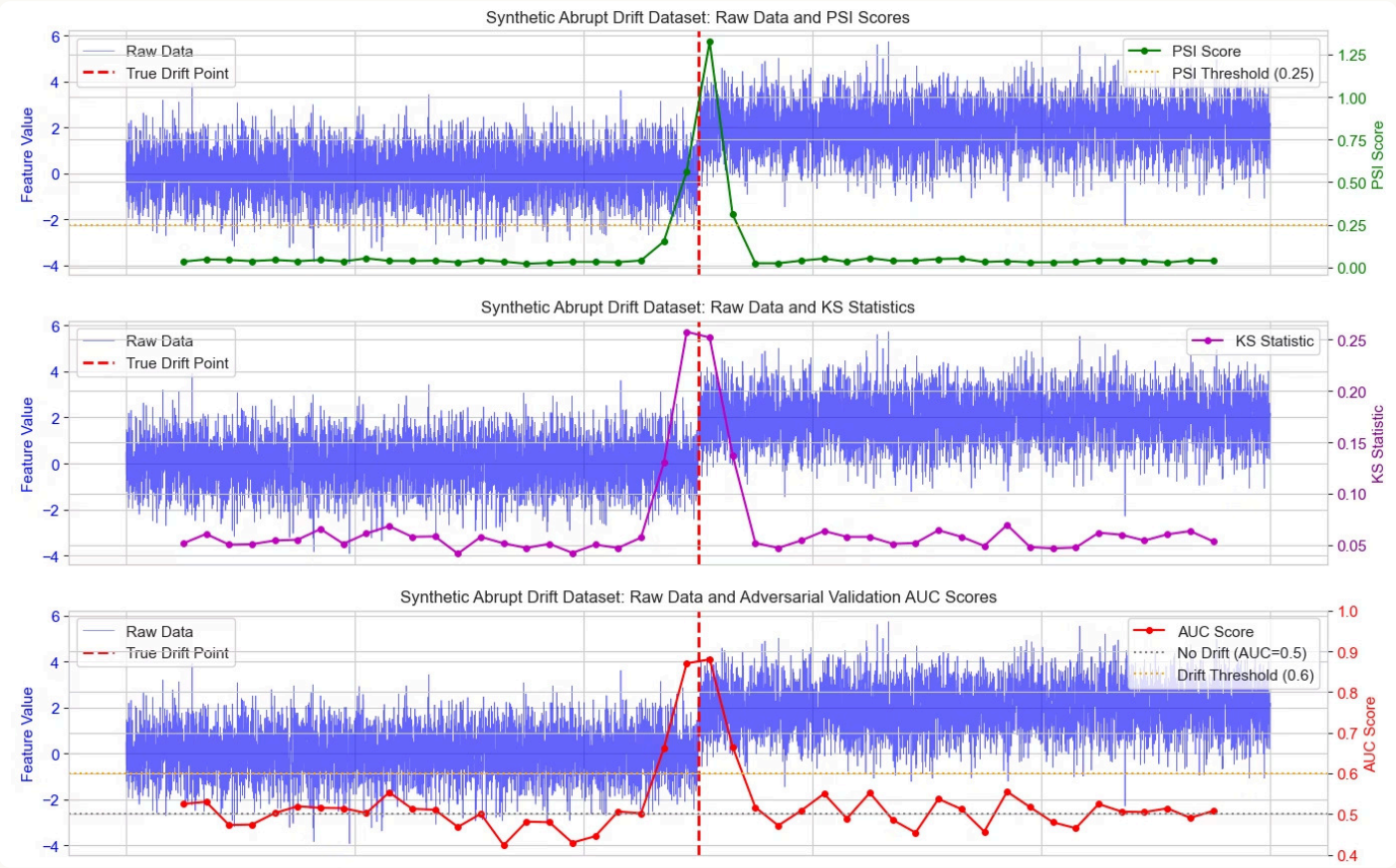


Figure: Time-Series visualization of Abrupt Drift

Experiment 1B: Gradual Drift (Synthetic Dataset)

Experiment 1B: Gradual Drift (Synthetic Dataset)

This experiment evaluates adversarial validation's effectiveness in detecting subtle, ongoing shifts in data distribution. We simulate a linear change in all features to observe if the method can capture this gradual drift.

Experiment Configuration:

drift_point	5000
n_samples	10000
n_features	5
AV classifier	LogisticRegression + StandardScaler
class_weight	balanced
CV folds	5
Permutation B	200
alpha	0.05

Streaming window rule:
W_ref = X[t - 500 : t], W_cur = X[t : t + 500]
start t = 500, step = 200

Experiment 1B: Gradual Drift (Synthetic Dataset)

The statistical evidence below confirms the engineered gradual drift across all features.

Feature	Before Drift (Mean +/- Std)	After Drift (Mean +/- Std)	Drift?
Feature_0	0.4150 +/- 1.0493	1.6043 +/- 1.0422	YES
Feature_1	0.4264 +/- 1.0335	1.6047 +/- 1.0232	YES
Feature_2	0.3794 +/- 1.0272	1.5859 +/- 1.0382	YES
Feature_3	0.3907 +/- 1.0314	1.6059 +/- 1.0243	YES
Feature_4	0.4165 +/- 1.0076	1.5961 +/- 1.0215	YES

The table clearly shows that all features exhibit a gradual increase in their mean values, moving from approximately 0.4 to 1.6. This aligns with the expected behavior of the synthetic gradual drift, where features were designed to increase linearly over time.

Results

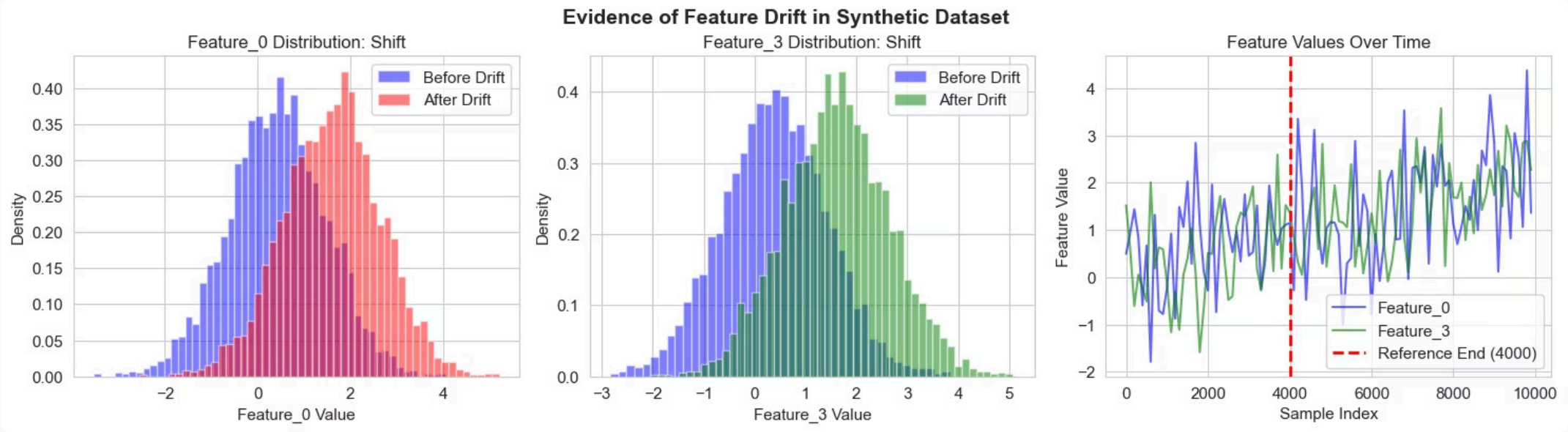


Figure: Histogram and Time-Series visualization of Gradual Drift

Results



Figure: Time Series visualization of Gradual Drift

RESEARCH FOCUS

Research Goals & Questions

1

Research Goal:

Investigate adversarial validation as a feature drift detection technique

2

Research Questions:

1. How effective is AV for detecting different drift types?
2. Can AV detect drift earlier than statistical/error-based methods?
3. How does AV compare to PSI, KS, DDM, ADWIN on benchmark datasets?

3

Expected Contributions:

1. Formal AV framework for unsupervised feature drift detection
2. Systematic benchmark evaluation across synthetic & real datasets
3. Empirical comparison with established drift detection methods
4. Analysis of detection delay (lead time) and stability

Research Plan

December 2025
Background reading, finalize topic

January 2026
Proposal preparation & submission (3rd week)

February 2026
Begin experiments

March 2026
Continue experiments, evaluate drift methods

April 2026
Write & finalize conference paper, submit paper

May 2026
Thesis defense (1st & 3rd week)

June 2026
Corrections after defense

July 2026
Graduation processing

CITED

References

Full reference list available in written proposal. Key sources:

Foundational Concepts:

- Sculley et al. (2015) - Hidden Technical Debt in Machine Learning Systems, NeurIPS
- Gama et al. (2014) - A Survey on Concept Drift Adaptation, ACM Computing Surveys
- Breck et al. (2017) - The ML Test Score: A Rubric for ML Production Readiness, Google Research

Drift Detection Methods:

- Bifet & Gavaldà (2007) - Learning from Time-Changing Data with Adaptive Windowing (ADWIN)
- Cobb & Van Looveren (2022) - Context-Aware Drift Detection, ICML

Adversarial Validation:

- Pan et al. (2020) - Adversarial Validation Approach to Concept Drift (Uber MaLTA), arXiv
- Raeder et al. (2012) - Managing Dataset Shift by Adversarial Validation for Credit Scoring

Benchmark Framework:

- Lu et al. (2024/2025) - Early Concept Drift Detection via Prediction Uncertainty (PU-index/PUDD), AAAI